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Predictive model of algal biofuel production based on experimental data

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ABSTRACT

Algal biofuels are of growing interest in the quest to reduce carbon emissions in the atmosphere but the sensitivity of the fuel production to various factors is not well understood. Therefore, the effects of temperature, light intensity, carbon concentration, aeration rate, pH, and time on the CO_2 biofixation rate of *Chlorella vulgaris (ISC-23)* were investigated using experimental, and Genetic Programming (GP) modeling techniques. The impacts of applying the cement industrial flue gas as a source of carbon, useful for the growth of microalgae, were also studied. *Chlorella vulgaris* (ISC-23) was cultivated in a laboratory photobioreactor on a BG-11 medium. The developed GP model was used to optimize the CO_2 biofixation based on the studied variables and produce a predictive equation. By using statistical measurements and error analysis, the predictive equation was shown to agree with the experimentally obtained values. It was found that the optimum conditions occur at 26° C, and 3200 lx of light, in the existence of CO_2. Applying 6% CO_2 as the input with the aeration rate of 0.5 vvm in 11 days was also reported as the optimum scenario for algae production with keeping the pH close to 7.5. The results indicate that the predictions determined with the proposed equation can be of practical worth for researchers and experts in the biofuel industry.

1. Introduction

Biofixation of CO_2 with algae is a promising approach to carbon capture and recycling with the side benefits of valuable biofuel and byproduct production [1][14]. Inorganic carbon could be assimilated into organic matter through marine and freshwater photosynthesis, which is an indication of fluxes on the planet with the capacity of absorbing about one-third of anthropomorphic atmospheric CO_2 [2]. In comparison to other systems, microalgae cultivation is a new system that has several advantages, such as higher oil yield and growth rate, as well as the potential to produce byproducts [3,4][47].

The cement sectors, which propagate 6% of global CO_2 emissions, have tried to mitigate the energy consumption provided by petroleum-based fuels [5]. However, it has not been successful in CO_2 reduction during the calcination process, thereby making carbon capture technologies essential. Additionally, other pollutants containing toxic gases and heavy metals are also produced during the cement-making process [6][42][43][44]. These constituents inhibit algae production [8–10], however they can be useful on some algae species [8,11,12]. Emissions from the cement industry or power plants have high concentrations of carbon dioxide and some nutrients useful for the cultivation of algae [9].

The algae productivity for CO_2 biofixation has been reported in many studies [13]. However, there is a lack of resources considering a trade-off between optimal growth and CO_2 biofixation rates. Among the various variables involved, temperature, light intensity, carbon concentration, aeration rate, pH, and time are the most important factors in algae biomass growth and CO_2 biofixation. Empirical methods of production prediction are expensive and time-consuming with some potential errors. Computational techniques may also lead to errors due to simplifying assumptions [45,46][48]. For successful and sustainable algal biofuel production, a robust mathematical model could obtain the optimum results for different growth conditions. The results could be validated with experimental data, promoting the optimization of the process to make it more profitable.

Many studies have addressed dynamic behavior modeling of algae and tried to identify algae growth rate, activation energy, and carbon biofixation with the application of artificial neutral networks (ANN), kinetic modeling [15,16], and Gaussian regression [17]. The effects of temperature on the growth of *Chlorella Kessleri* using Monod and

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Arrhenius kinetics have also been investigated [18]. However, a robust model integrating impacts of temperature, light intensity, carbon concentration, aeration rate, pH, and time on the CO_2 biofixation rate during the algae growth has not been studied so far in the literature. *Chlorella vulgaris* was selected as representative of fresh water algae. It has been widely used in the literature because of its high oil yield, salinity tolerance and energy content [19,20].

Genetic Programming (GP) as a subset of genetic algorithms that are based on principles of genetics, and has been widely applied in computer-based pattern recognition systems and nonlinear behavioral model development. To combine GP with regression in parameter estimation, Multi-Gene Genetic Programming (MGGP) has been shown to be a promising technique that uses population-based multiple sets of genes with a weighted linear combination of low-order non-linear transformations of the input variables. Identifying the input and output data for training and testing the model, GP is capable of generating a robust model to formulate an equation used for forecasting [21]. A GP works by starting with a population size comprised of "chromosomes" containing "cells" that represent a possible set of input variable values. Cells are exchanged between "parent" chromosomes to form "offspring" chromosomes. The offspring are tested for "fitness" and the fittest offspring are kept in the "gene pool" and allowed to procreate for the next "generation." The procreation continues for a certain number of generations. Periodically a "mutation" gene or "merged independent run" is introduced to search a different location of solution space. The concept of "tree depth" from graph theory allows for characterizing the size and extent of the GP algorithm solutions considered.

The purpose of this study, therefore, is to define a prediction model of carbon biofixation rate using GP produced by a photobioreactor (PBR) process on *Chlorella vulgaris* algae under different fixed input values of CO_2, while also taking into consideration the effects of a simulated gas component of cement industrial flue gas as the source of carbon. This is accomplished by taking experimental data to train the predictive model.

2. Materials and methods

To obtain a predictive model, first experimental results were obtained to train the model.

2.1. Experimental details

2.1.1. Flue gas analysis, experimental setup and conditions

The appropriate microalgae species for biogas production must have (1) high CO_2 utilization capacity and limited nutrient requirements; (2) high lipid productivity; and (3) high-value biomass co-products. *Chlorella* appears to be a suitable candidate based on previous studies [22,23][7]. Therefore *C. vulgaris ISC-23* was selected, and obtained from the Iranian Academic Center for Education, Culture & Research at the Beheshti University.

In this study, along with the different fixed input values of CO_2 that were taken into consideration, the effects of a simulated gas component of cement industrial flue gas were also investigated to evaluate the adaption strategy of a new aeration system on algae. A cement gaseous aeration system would help the environment by reducing the CO_2 emissions and producing alternative energy along with some other valuable by-products. Each experiment contains 3 replicate sets of data on each biomass production scenario, during the sampling process, to provide confidence in the data. The experimental method has also been validated on the base case scenario, with an abundant and well-known biomass sample as a control. Finally, calibration, instrumentation or procedures have been verified with internal standards.

The medium was prepared in a 1000-mL Erlenmeyer flask, autoclaved at 120 °C, 1 bar, over 20 min with distilled water enriched with BG11 [24]. BG11 media consisting of: $NaNO_3$ (1500 mg/L), K_2HPO_4-$3H_2O$ (4 mg/L), $MgSO_4$-$7H_2O$ (75 mg/L), $CaCl_2$-$2H_2O$ (27 mg/L), citric acid (6 mg/L), Ammonium ferric citrate (6 mg/L), EDTANa_2Mg (1 mg/L), Na_2CO_3 (20 mg/L), H_3BO_3 (2.860 mg/L), $MnCl_2$-$4H_2O$ (1.810 mg/L), $ZnSO_4$-$7H_2O$ (0.222 mg/L), Na_2MoO_4-$2H_2O$ (0.390 mg/L), $CuSO_4$-$5H_2O$ (0.079 mg/L), and $Co(NO_3)2$–$6\ H_2O$ (0.0494 mg/L) [25]. To estimate the capability of the microalgae to capture CO_2 as a nutrient resource, the cycle plan was tested in a BG11 medium without main C– (BG11-C). Consequently, Na_2CO_3 was eliminated from the BG11 recipe. Also, to avoid any precipitate, K_2HPO_4-$3H_2O$ was sterilized in a different Erlenmeyer flask. In all tests, the pH was set to 7.4 (for optimum results) with the application of HCL and NaOH, except for one experiment which studied the impacts of pH on the algae growth. All compounds used were analyzed and procured from Merck, Inc.

In the experimental microalgae cultivation system, the strain was placed in three replicate Erlenmeyer flasks (1000-mL) with an 800-mL working volume. Enriched flasks were placed at room temperature (27 ± 2 °C), centrifuged at 100 rpm, and the cultures were grown under continuous light intensity (3600 ± 200 Lux) provided by three fluorescent lamps [26].

To reduce lag and to aid in microorganism adaptation to CO_2, all flasks except the control sample were exposed to normal room air containing 1% CO_2 for seven days before inoculation. All cultures were inoculated with a similar initial cell concentration of 2.47×10^6 cells mL^{-1} (0.1 OD680). Microalgae has chlorophyll and typically 680 nm is associated with chlorophyll absorption through the suspension of cells. pH was also adjusted to 7.4, after which 120 g L^{-1} of NaOH was added to avoid severe pH changes.

After four days inoculum flasks were transferred to a 1-L glass bubble column photobioreactor (PBR) in a biosafety cabinet. The bubble column was closed by placing a stopper with two orifices, one for the inlet gas flow and the second for sample collecting and outlet gas flow at the top. Parafilm foil was placed to seal it. Air stone sparging was used to bubble the gas containing different CO_2 concentrations (4, 6, and 8% *v*/v) and by synthetic cement flue gas (SCFG). The flue gas analysis was done by a portable Combustion gas analyzer (KIMO KIGAZ 310) based on ISO 9096:2017 standards from a clinker kiln stack in the Ilam cement plant located in Ilam, Iran. The flue gas was composed of NO_2 (13.71 mg/Nm^3), CO_2 (4.5%(*v*/v)), NO (153.28 mg/Nm^3), CO (49.39 mg/Nm^3) and SO_2 (2.82 mg/Nm^3), dust (75.91 mg/Nm^3), and oxygen (12.5% (v/v)). Natural gas was used as the main fuel and the exhausted gas temperature was 107.4 °C [26]. Median pH was monitored every ten minutes before, during, and after each bubbling process. All flasks were oriented vertically and received light from three sides with a 16:8 h light/dark photoperiod under 3200 ± 200 Lux light intensity. The optical path of the PBR was 4 cm.

Different CO_2 concentrations were mixed in the air at 0.25 vvm with 3 florescent light bulbs under the temperature of 26 °C. The concentrations of 4%, 6%, and 8% CO_2 were provided and applied for up to 8 days. Every 8 h, the optical density, dry weight, and pH were sampled [23,27–30].

The quality control (QC) on replicate measurements was done with a limit of 10% relative variability between them. The relative standard deviation was used as a precision metric. Two researchers worked on two different days on the total of 6 replicates with the same material, following the standard procedure for control purposes. The precision of the experiments was checked to have a relative error of less than ± 5%, based on interquartile ranges of data, which led to defining the outlier data. In this regard, the points that fell outside of the 1.5 × interquartile range were defined. One outlier was detected during the QC, and it was removed from the analytical calculations to meet our target precision.

2.1.2. Characterizing parameters and biochemical analyses

To optimize the algae growth, the behavior of algae during the first 19 days of cultivation was studied. First, the temperature was assumed constant at 26° C. The algae were aerated with ambient air under two 50-watt yellow light bulbs in the first 10 days. Then, the concentration

of CO_2 was changed to 6%. On the 12th day, an additional florescent bulb was installed parallel to the photobioreactor. Finally, on the 14th day, the lighting system was changed to 3 florescent bulbs located with an angle of 120 degrees to the bioreactor, providing 3200 lx.

Furthermore, the cell density evaluation was performed. In addition to cell counting, an assessment of the optical density and specific growth rate are highly dependent on growth conditions in the medium. Physiological parameters were measured every 10 days by collecting culture samples from the PBR system at the same time every day. Cultures were manually mixed before sampling. Based on the observations, the microalgae cell counting was performed daily in the microalgae 1 mL suspension samples. This was performed through cell counts using an Improved Neubauer hemocytometer under an optical microscope. To obtain the biomass concentration (dry weight per volume) of the cells, optical density at 680 nm (OD680 nm) was measured by UV–Vis spectrophotometer, and an absorbance range of 0.1–0.5 was applied by dilution in optical densities that are > 1.0 [31].

Three samples of 10 mL were centrifuged (5000 rpm, 10 min). The extra liquid was removed and transferred to a 1.5 mL tube, washed twice with distilled water, then centrifuged at 13,000 rpm for 10 min, and finally moved to microtubes and were heated at 80 °C for 1 day and weighed.

For the measurements of the specific growth rate and productivity, cell concentration values were observed using Eq. (1)

$$\mu = \frac{\ln N_2 - \ln N_1}{t_2 - t_1}, \tag{1}$$

where μ is specific growth rate, and N_2 and N_1 indicate cell concentration at different times of t_1 and t_2 (days). Average biomass productivities (P, gdw L^{-1} d^{-1}) were defined by Eq. (2)

$$P = \frac{X_t - X_0}{t_t - t_0}, \tag{2}$$

where X_t and X_0 correspond to biomass concentration (in gdw L^{-1}) at the end (t_2) and start of cultivation (t_0), respectively [32–34].

The carbon dioxide biofixation rate is another important factor to be investigated. The biofixation of carbon with algae helps to convert the polluted flue gases of large-scale industries into biofuels. The carbon dioxide fixation rate (R_c, gcL^{-1} d^{-1}) depends on carbon content and productivity, and was determined by Eq. (3):

$$R_C = C_C P\left(\frac{M_{CO_2}}{M_C}\right), \tag{3}$$

where, C_c is algae concentration, p is the microalgae biomass production rate and M is the molar mass. A Shimadzu TOC analyzer was used to define the biomass carbon content. The average C_C (wt) for different CO_2 concentrations was found to be 49.2% for *Chlorella vulgaris (ISC-23)* [34].

2.2. Mathematical model and validation

To develop a mathematical model for algal biofuel growth GP has been used based on the experimental data described earlier. The Genetic Programming and Symbolic Data Mining Platform (GPTIPS 2), a free symbolic data mining platform and gene-centric method, has been applied by using a biological machine learning approach based on genetic programming to simulate the production process. The method has been introduced by Searson et al., in 2015 [35]. GPTIPS applies MGGP algorithm to generate trees, referred to as multigene symbolic regression. If successful, the GPTIPS not only finds hidden and nonlinear data connections, but also creates accurate equations predicting the model behavior. The data-driven models could be built even with noisy, highly correlated, and superfluous input variables. The tool is a biologically inspired method applying machine learning by random generation of a population of programs, then mutating the best performing trees towards a new population. It moves forward to the point where the population contains programs with solved tasks. GP as a symbolic regression creates an empirical mathematical model of data. GP has the advantage of evolving both the structure and parameters, unlike traditional regression models [35]. The complex model portfolio is generated based on the experimental data generated by the authors. For the model training purposes here, 70% of the data have been used, and the rest have been applied for model validation.

Independent variables in this study are temperature, light intensity, carbon concentration, aeration rate, pH, and time. The general form of the model is presented in Eq. (4):

$$R_f = f(X_1, X_2, X_3, X_4, X_5, X_6), \tag{4}$$

where R_f is the carbon biofixation rate, and Xi represents different variables (temperature, light intensity, carbon concentration, aeration rate, pH, and time, respectively).

To validate the model, the experimental and modeling results were evaluated. The data were divided into training data and test data. Methods used for comparison are the Root Mean Square Error (RMSE), Sum of Squared Errors (SSE), Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2. Multigene symbolic regression was used, which can find the best fit for a given dataset. The exploratory Post Run Analyses, Multigene Genetic Programming yields several exploratory visualization tools to analyze the results. The population browser using the “popbrowser” function was used to illustrate the population predictive performance.

The “runtree” function was used to provide plots comparing the predicted and measured values [35]. The acceptable range for each independent variable was based on previous studies (Table 1).

3. Results and discussion

The predictive model is based on the observations in the results of experimental setup, which predicts growth conditions including the impacts of temperature, light intensity, carbon concentration, aeration rate, pH, and time on the *Chlorella vulgaris* (ISC-23) culture.

3.1. Experimental results

3.1.1. Growth profile of algae

First, the correlation between optical density vs dry weight and cell number were determined (Fig. 1). To define the cell number, the Neubauer-improved cell counting chamber was used. Based on the results, the obtained equation for the dry weight, and the cell concentration are:

$$\textit{Chlorella Vulgaris}\ \text{dry weight} = 0.2749\,(OD_{680}) + 0.0026 \tag{5}$$

and

$$\textit{Chlorella Vulgaris}\ \text{number of cells} * 10^7 = 2.2189\,(OD_{680}) + 0.1614 \tag{6}$$

where R^2 equals 0.992 and 0.996 for the dry weight and the cell numbers, respectively.

The effects of using the cement flue gases on the algae growth are reported in Fig. 2. A 5.88% decrease in the average cell number, cell/L, was obtained for Treatment B which is the algae being aerated in two stages: standard conditions in the first 8 days, and aeration with

Table 1
Acceptable ranges of each variable for the *Chlorella vulgaris* (ISC-23).

Variables	Range	Ref.
Temperature (C)	10–35	[36]
Light intensity (μ. Mol. photon. m^{-2}. s^{-1})	40–470	[37]
Carbon concentration (%)	4–8	[38]
Aeration rate (vvm)	0.1–1	[39]
pH	6–10	[39]
Time (day)	< 38	[40]

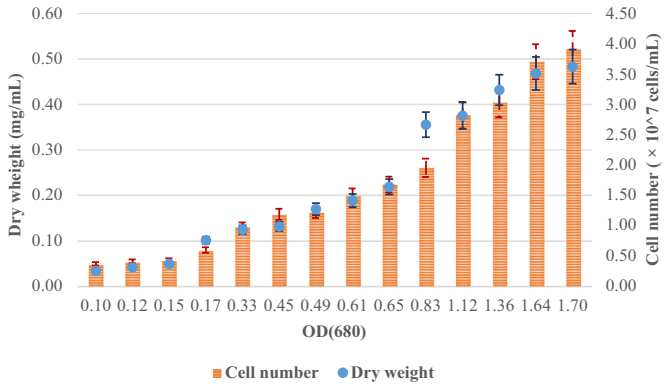


Fig. 1. The effect of optical density on the *Chlorella vulgaris*' dry weight and cell number. Results are presented in mg dry weight per ml, and cell concentration ($\times 10^7$) per mL, and expressed as the mean ± SD (n = 3).

cement-making exhaust gas after the 8th day. Treatment A consists of the algae being cultivated in the standard condition. The results show similar growth curves for the two cultures which demonstrate minimal impacts of the simulated flue gases application.

3.1.2. Algae growth optimization

Results concerning the optical density (OD) and the specific growth rate over the 19 days of algae cultivation confirmed a regular pattern during the first 10 days for the OD and the specific growth rate. After 10 days, the addition of CO_2 to 6% increased the specific growth rate, thereby improving the system efficiency. Changing the lighting system after the 12th day shows an increase in the rate. Increasing the florescent bulbs also positively affected the growth.

3.1.2.1. The effects of CO_2 on algae growth. Fig. 3 presents the experimental results. During the four rounds of experiments applying 4%, 6%, and 8% of CO_2, along with the utilization of cement-making emitted flue gases, the amounts of Optical Density (Fig. 4A), specific growth rate (Fig. 4B), productivity (Fig. 4C), and CO2 biofixation rate (Fig. 4D) have been determined, and plotted.

The results show that during the first test with 4% contribution of CO_2, the maximum amount of algae growth was between the 1st and 4th day. Also, the most amount of biofixation rate occurred on the 6th

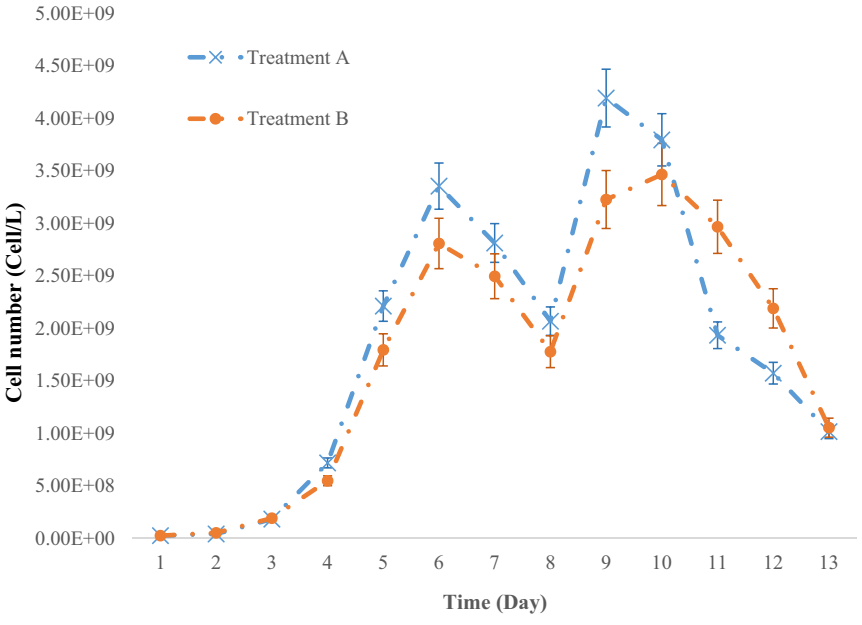


Fig. 2. The effect of exhaust gas on *Chlorella vulgaris* (ISC-23) cell number. (Treatment A algae was cultivated in the standard condition, and in Treatment B, the aeration was done in 2 stages: standard conditions first and adding cement-making exhaust gas after the 8th day.) Results are presented in cell number per L, and expressed as the mean ± SD (n = 3).

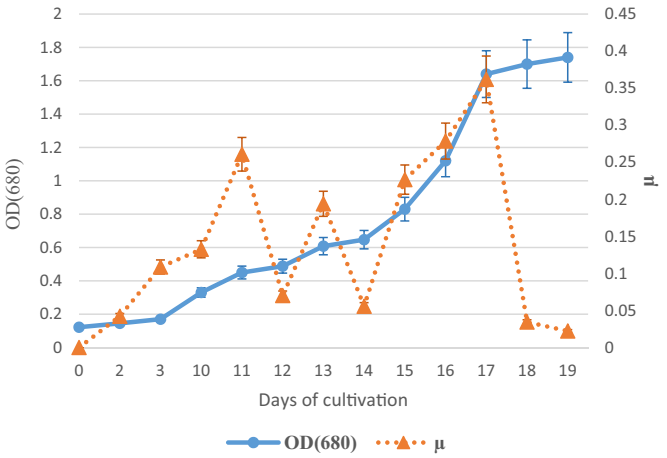


Fig. 3. Optical Density (OD) and specific growth rate of algae with the application of 6% CO_2 versus culturing days Results are presented in lumen for light intensity, and specific growth rate per day, and expressed as the mean ± SD (n = 3).

day of production. The percentage of CO_2 was also increased, which exhibited a dark green color of algae biomass after the 6th day. The darkness adversely impacted the light absorption which led to less efficiency in the production rate. This period could be the best time for algae cultivation.

The results of the second round (6% CO_2) showed that the specific growth rate was maximum at the 1st day. The amount of optical density (OD) was equal to 7.52 after 10 days which showed the effectiveness of this setup. The specific growth rate (μ) was also equal to 1.21 which showed an ideal condition for distribution and aeration.

The 3rd round of tests with the application of 8% CO_2 showed a linear change of OD over time as expected. After 5 days the dark algae color in the medium decreased the cultivation efficiency (OD = 3.5 nm). The utilization of cement gas also reported that OD reached 5.8 nm after 11 days, which interestingly is higher than the utilization of 4% and 8% of CO_2.

The experimental setup is also responsible for the variations of pH over time in the 8% CO_2 scenario (Fig. 5). The results show the maximum slope occurred in the maximum special growth rate. Also, the decrease in pH reported after the 7th day is due to lack of enough light

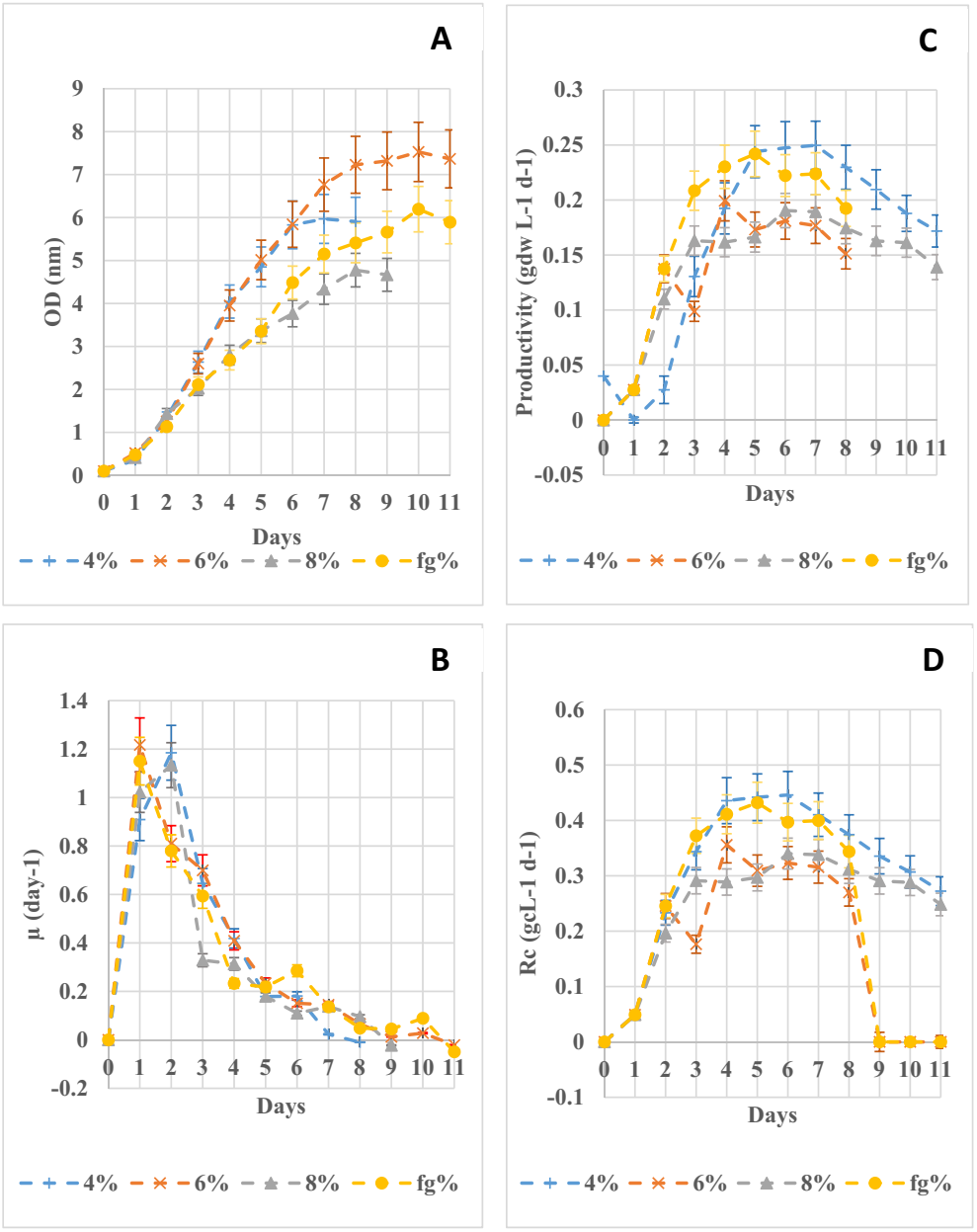


Fig. 4. The effects of CO_2 utilization percentage on Optical Density (OD) (A), specific growth rate (B), productivity (C), and CO_2 biofixation rate (Rc) (D) over time with additions of 4%, 6% and 8% CO_2, and cement-making emissions. Results are presented in Light intensity (μ. Mol. photon. m-2. s-1), specific growth rate per day, productivity per liter per day and CO_2 biofixation rate per day, and expressed as the mean ± SD (n = 3).

in the system.

Furthermore, the maximum growth occurred with 6% CO_2 of 1.21 day^{-1}, followed by the 4% CO_2 scenario with a rate of 1.18 day^{-1}. Since CO_2 is the only variable in this test, having a high access to carbon, risk of cell poisoning due to a reduction in pH, lower initial inoculation, and the effectiveness in utilizing enough CO_2, and the biochemical behavior of *Chlorella* with a high dose of carbon could account for reduced efficiency in scenarios with > 6% CO_2. Although using the cement flue gases with *Chlorella vulgaris (ISC-23)* has proven to be a good solution [41], the existence of SO_2 would have some adverse impacts on the growth rate due to changing pH. Nitrogen would have the same result, but with lower impacts, since it could be used to feed the algae based on the amount of oxygen available.

Comparing the results of the CO_2 biofixation rate from cultivated microalgae for different inputs of CO_2 with the aeration of 0.25 vvm showed that the best results were seen in the 6% CO_2 mixture. After 6 days of growth, the peak CO_2 biofixation reached 0.445 g. l^{-1}. d^{-1} for the application of 6% CO_2 in the mixture, giving an estimated OD of 5.78 nm. The decrease reported after the 7th day is likely due to increased OD and decreased light intensity, suggesting an optimal OD value of 5 nm.

The results of maximum biomass concentration (Xmax) in different CO_2 contributions, an indication of cell density, showed that the 4% application of CO_2 results in 8.33 $mg.ml^{-1}$. However, the 6% contribution of CO_2 showed the most promising results where up to 14.06 $mg.ml^{-1}$ of *Chlorella vulgaris*. Increasing CO_2 to 8% reduced Xmax to 7.92 $mg.ml^{-1}$, whereas the application of cement production exhaust gas grew 11.8 mg ml^{-1} of algae.

3.1.2.2. The effects of aeration rate on growth. Table 2 presents the results of the maximum biomass concentration (X_{max}), maximum production rate (P_{max}), and maximal CO_2 biofixation rate for different

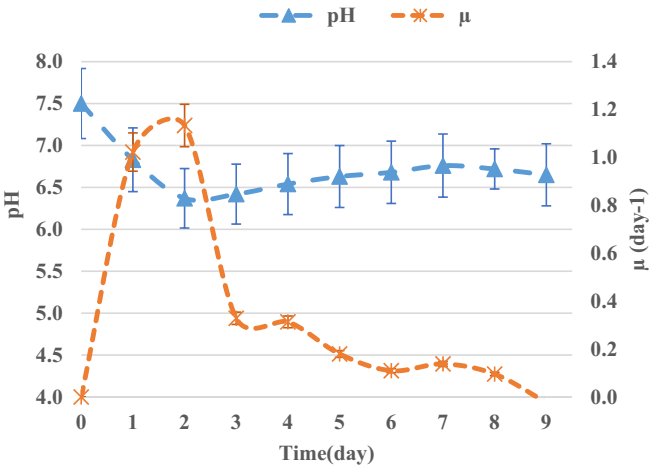


Fig. 5. Comparison of pH variations versus specific growth rate over time (with 8% CO_2 input). Results are presented in specific growth rate per day, and pH, and expressed as the mean ± SD (n = 3).

Table 2
The maximum biomass concentration (Xmax), maximum production rate (Pmax), and maximal CO_2 biofixation rate in different aeration rates. Results expressed as the mean ± SD (n = 3).

Aeration rate (vvm)	Xmax ($g.L^{-1}$)	Pmax ($g{\cdot}L^{-1}{\cdot}d^{-1}$)	Maximal CO_2 biofixation rate ($g{\cdot}d^{-1}$)
0.10	9.195	1.157	2.066
0.25	9.917	1.240	2.214
0.50	14.294	1.804	3.222
0.75	12.054	1.575	2.812
1.00	11.342	1.459	2.605

aeration rates with the assumption of having only 6% CO_2 cultivated in 8 days. The 0.5 vvm value shows the most promising results in all the metrics. Increasing the aeration rate would increase CO_2 in the medium, and pH would decrease the algae cultivation efficiency at first and increased turbulence in the final days.

The results show that having a temperature of 26° C and 3200 lx of light in the existence of CO_2 to get the highest biofixation rates, and the reduction of fatty acids are recommended. Applying 6% CO_2 with an aeration rate of 0.5 vvm was also reported as the most efficient scenario for algae production. 11 days of continuous production with the pH close to 7.5 led to 6.58 of OD. Also using fresh water shows higher efficiency compared to sea water. The results show the capability of receiving high amounts of CO_2 in *Chlorella vulgaris (ISC-23)*, leading to lower CO_2 due to the biofixation process. Furthermore, the impacts of applying the cement production flue gases agreed with other pre-defined scenarios.

The entire dataset applied in this study passes our target precision of 10% after the removal of one outlier. The dataset used for the GP modeling falls within ± 1 standard deviations of the mean of the consensus data.

3.2. GP model development and validation

To develop a predictive model, a GP was employed to find empirically-derived equations from the experimental data. The input parameters for the implementation of GP are presented in Table 3. The values of population size and maximum number of generations required depend on the data complexity. The input variables in this study are the total of 6 including temperature, light intensity, carbon concentration, aeration rate, pH, and time. The maximum numbers of genes and depth of genes were 8 and 4, respectively. The maximum number of genes ranges between 4 and 10 in the literature, and a high number could lead

Table 3
GP parameter values.

Run parameter	Value
Population size	500
Max. generations	150
Generations elapsed	96
Input variables	6
Training instances	58
Tournament size	25
Merged independent runs	5
Run timeout (*sec*)	60
Max. genes	8
Max. tree depth	4

to a better approximation, but there is always a risk of overfitting. For each data set, the initial 70% of the population size is allocated to training (training instances), and the remaining 30% to testing (tournament size). Each run has been forced to terminate in 60 s to optimize the code for faster execution of output [35].

First, 2500 iterations were run, and the correlations and the residual, or errors were determined. Using the experimental data, the following biofixation rate equation was found to have the best fitness value (Eq. (7)).

$$\begin{aligned} CO_2\ \text{biofixation} = {} & 4.73\,x_2 - 5.05\,x_3 + 92\,x_5 - 0.0116\,x_1^3\,x_5^2 - 4.36x_2 \\ & x_5^2 + 0.105\,x_2\,x_3\,x_5^2 + 0.0311\,x_1\,x_5^2 \\ & x_6\,(x_4 + x_6) - 0.0343\,x_2\,x_3\,x_5\,(2\,\tanh(x_5) + 1) - 104 \end{aligned} \quad (7)$$

where x_1 is growth time (days), x_2 is temperature (C), x_3 is light intensity (μ. Mol. photon. m^{-2}. s^{-1}), x_4 is CO_2 concentration (%), x_5 is aeration rate (vvm), and x_6 is pH. Different obtained equations with the lowest R^2 have also been reported in Table 4.

Applying the population browser (popbrowser) function to illustrate the population predictive performance, $(1 - R^2)$ vs. expressional complexity has been plotted (Fig. 6). The green, blue and red dots in the figure represent the Pareto front, non-pareto, and the best model, respectively. The green dots are representative of the satisfied results.

Furthermore, 8 genes have been defined (Table 5). More genes would help increase the accuracy of the model, but also increase the running time and complexity of the model. The weighting results showed that the Genes 1, 3, and 6, have the highest contributions on the final model, respectively.

The evaluation of errors, comparing the results from the training data versus the test data is in Table 6. Goodness of fit (R^2), root mean square error (RMSE), mean absolute error (MAE), sum of squared errors (SSE), maximum absolute error (MAE), and mean squared error (MSE) were used for the model evaluation. The results show a 94% R^2.

Fig. 7 plots the proposed GP model results versus the actual experimental data. High values of correlation are found to be 0.879 and 0.939 for the training and test data, respectively. Additionally, the accumulation of data points is located very close to the reference line of 45° line or perfect agreement, which shoes how accurate the GP model is.

4. Summary and conclusions

GP was implemented to obtain a CO_2 biofixation rate model dependent on temperature, light intensity, carbon concentration, aeration rate, pH, and time. For this purpose, experimental data were collected in the laboratory cultivating *Chlorella vulgaris (ISC-23)* under different conditions. The effects of different combinations of CO_2 input, aeration rate, application of simulated flue gases from a cement-making plant were studied as individual growth scenarios. At the end the GP modeling results were evaluated with a comprehensive statistical analysis.

The results recommend having a temperature of 26° C and 3200 lx

Table 4
Proposed biofixation rate equations, ranked based on experimental data fitness, where x_1 is growth time (days), x_2 is temperature (C), x_3 is light intensity (μ. Mol. photon. m^{-2}. s^{-1}), x_4 is CO_2 concentration (%), X_5 is aeration rate (vvm), and x_6 is pH.

Model ID	Goodness of fit (R^2)	Model complexity	Model
151	0.879	98	$4.73 \times {}_2 - 0.55 \times {}_3 + 92.0 \times {}_5 - 0.0116 \times {}_1{}^3 x_5{}^2 - 4.36 \times {}_2 \times {}_5{}^2 + 0.105 \times {}_2 \times {}_3 x_5{}^2 + 0.0311 \times {}_1 \times {}_5{}^2 x_6 (x_4 + x_6) - 0.0343 \times {}_2 \times {}_3 x_5 (2.0 \tanh(x_5) + 1.0) - 104.0$
1537	0.817	35	$1.24 \times {}_2 - 0.124 \times {}_3 - 26.2 \times {}_5 + 43.1 \tanh(x_5) - 0.00292 \times {}_3{}^2 x_5 + 0.0134 \times {}_1 \times {}_3 x_5{}^2 - 28.3$
1506	0.814	32	$1.44 \times {}_2 - 0.171 \times {}_3 - 3.48 \times {}_5 + 0.065 \times {}_6 + 27.4 \tanh(x_5) + 0.158 \times {}_1 \times {}_5 + 0.00334 \times {}_3{}^2 x_5 - 0.0171 \times {}_2 \times {}_3 x_5 - 31.9$
1519	0.813	27	$1.36 \times {}_2 - 0.154 \times {}_3 - 11.8 \times {}_5 - 0.154 \tanh(x_2) + 27.5 \tanh(x_5) + 0.16 \times {}_1 \times {}_5 - 0.0023 \times {}_3{}^2 x_5 - 30.1$
602	0.766	20	$0.477 \times {}_2 + 22.7 \times {}_5 + 14.0 \tanh(x_5) + 0.201 \times {}_1 \times {}_5 - 1.18 \times {}_2 \times {}_5 - 13.7$
1501	0.678	17	$0.066 \times {}_2 + 0.066\ 3 + 3.28 \times 5 + 5.52 \tanh(x_5) - 0.0015 \times {}_3{}^2 x_5 - 5.83$
1577	0.415	15	$7.42e^{-4} \times {}_3{}^2 x5 - 0.0182 \times {}_3 + 0.879$
2028	0.105	2	$0.071 \times {}_1 + 0.0118 \times {}_2 - 0.224$

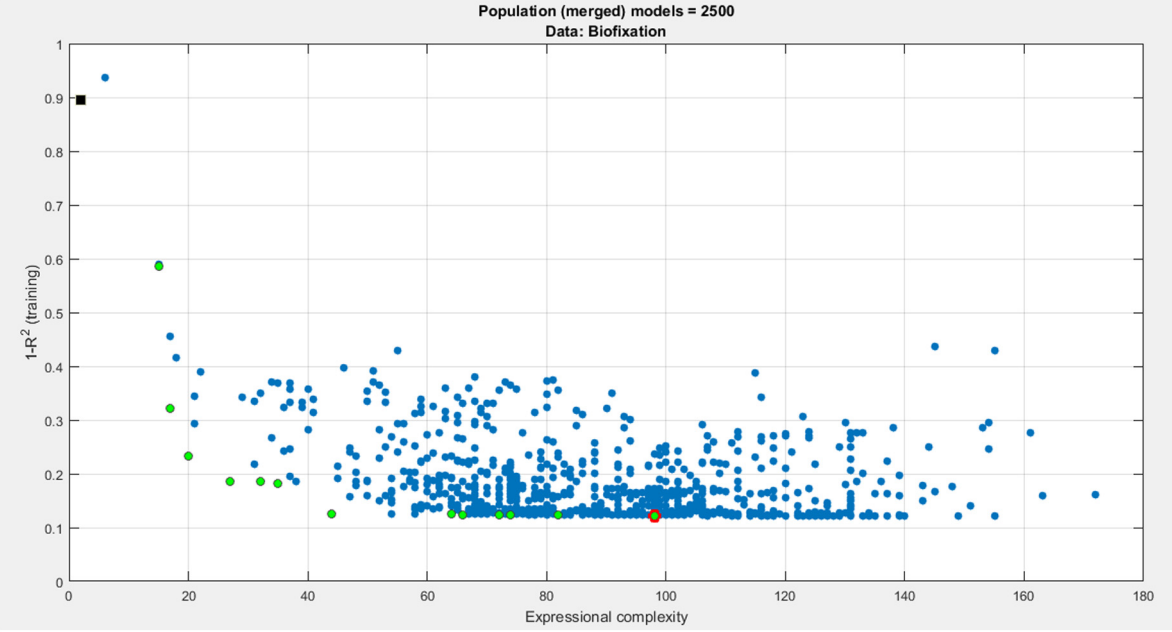


Fig. 6. Visualization of the regression model population using population browser.

Table 5
Genetic Programming (GP) genes.

Term	Value
Bias	-104.0
Gene 1	$-4.36 \times {}_2 \times {}_5{}^2$
Gene 2	$0.0311 \times {}_1 \times {}_5{}^2 x_6 (x_4 + x_6)$
Gene 3	$92.0 \times {}_5$
Gene 4	$0.55 \times {}_2 - 0.55 \times {}^3$
Gene 5	$0.105 \times {}_2 \times {}_3 x_5{}^2$
Gene 6	$4.18 \times {}_2$
Gene 7	$-0.0116 \times {}_1{}^3 x_5{}^2$
Gene 8	$-0.0343 \times {}_2 \times {}_3 x_5 (2.0 \tanh(x_5) + 1.0)$

Table 6
Statistical analysis of training data versus the test data (goodness of fit (R2), root mean square error (RMSE), mean absolute error (MAE), sum of squared errors (SSE), maximum absolute error (MAE), and mean squared error (MSE)

Metric	Value
Training data	
R^2	0.8791
RMSE	0.2373
MAE	0.0814
SSE	3.2686
Max. abs. Error	1.7100
MSE	0.0563
Test data	
R^2	0.9393
RMSE	0.0820
MAE	0.0542
SSE	0.1547
Max. abs. error	0.2287
MSE	0.0067

of light in the existence of CO_2 to get the highest biofixation rates, and the reduction of fatty acids. Applying 6% CO_2 with an aeration rate of 0.5 vvm was also reported as the most efficient scenario for algae production. 11 days of continuous production with the pH close to 7.5 led to 6.58 of OD. The results show the capability of receiving high amounts of CO_2 in *Chlorella vulgaris (ISC-23)*, leading to lower CO_2 due to the biofixation process. Furthermore, the impacts of applying the cement production flue gases agreed with other pre-defined scenarios.

The newly developed GP model formed here based on experimental laboratory results. Cultivation of *Chlorella vulgaris (ISC-23)* in the PBR system with the application of BG11 medium are pre-defined and essential assumptions of this study, resulted in a simple formula that predicted the experimental results within 93% for a wide range of variable values. Thus, the new model can be used as a platform for further investigations to optimize future applications of *Chlorella vulgaris* (ISC-23) in PBR cultivation systems based on different growth

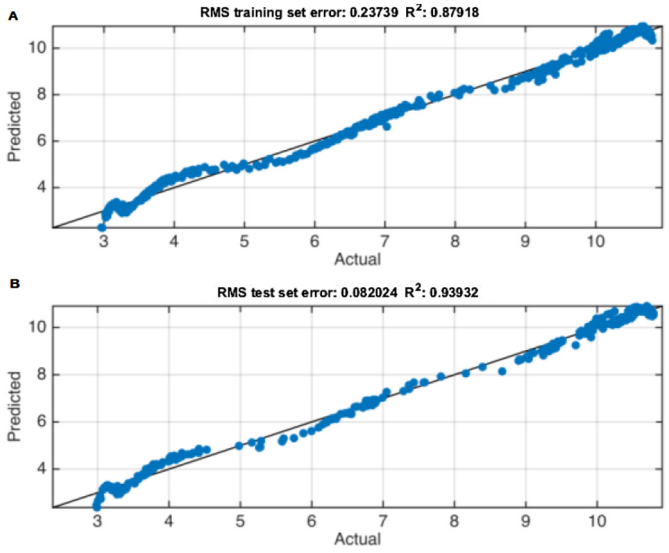


Fig. 7. Predicted values of GP model versus the experimental data with the reported goodness of fit (R2): (top) training dataset, (bottom) test dataset.

variables. However, for industrial projects, it is critical to develop a comprehensive model by performing a pilot scale study with the same assumptions based on the same kind of experimental data in this study, then scaling up to commercial scale, and finally, overseeing the limitations of the proposed predictive model in terms of comparing the significant input and output variables.

Statement of informed consent, human/animal rights

No conflicts, informed consent, or human or animal rights are applicable to this study.

Authors' agreement to authorship and submission

All the authors agreed to the authorship and submission of the manuscript to Algal Research.

Declaration of authors' contributions

Aryandokht Azari and Omid Bozorg haddad contributed to the design of the study, developing experimental procedure, tests, analysis, and interpretation of data.

Hossein Tavakoli and Brian Barkdoll contributed to the data acquisition, GP analysis, and drafting and revising the article.

CRediT authorship contribution statement

Aryandokht Azari: Conceptualization, Methodology, Software, Investigation, Writing - review & editing. **Hossein Tavakoli:** Conceptualization, Methodology, Software, Writing - original draft, Writing - review & editing. **Brian D. Barkdoll:** Validation, Writing - review & editing. **Omid Bozorg Haddad:** Supervision, Validation.

Declaration of competing interest

The authors declare no commercial or financial conflict of interest.

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